

Operational Intelligence for Complex Industrial Systems

We turn industrial problem-solving into reusable decision intelligence — created once, codified, and delivered across enterprise operations.

NVIDIA INCEPTION GRAND CHALLENGE • KOREA FINAL

omelet.ai • info@omelet.ai • Seoul, founded 2023



15

Enterprise customers
18 active projects

\$1.4M

Contracted
PoC + build • 2025–26

5

Paid production deployments
GS • Glovis • CAU Hospital + 2

20%+

Delivery cost reduction
Logistics routing

2wk→90sec

Scheduling time
Nurse rostering, CAU Hospital

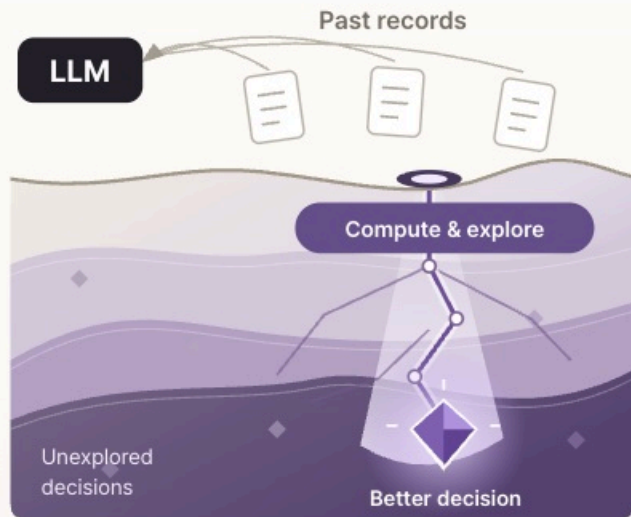
• THE CASE FOR OPERATIONAL INTELLIGENCE

Why Industry Needs a New Kind of AI

Go beyond past records. Find the right action. Carry the knowledge forward.

01 THE DATA GAP

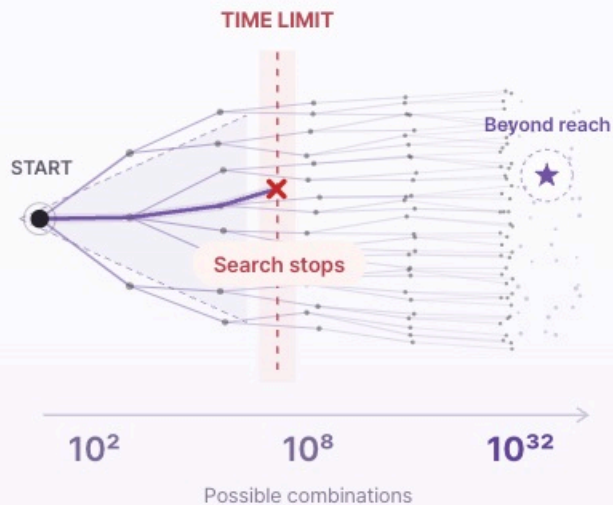
Better decisions must be discovered.



Logs tell us what happened.
Compute what should happen.

02 THE COMPUTATION GAP

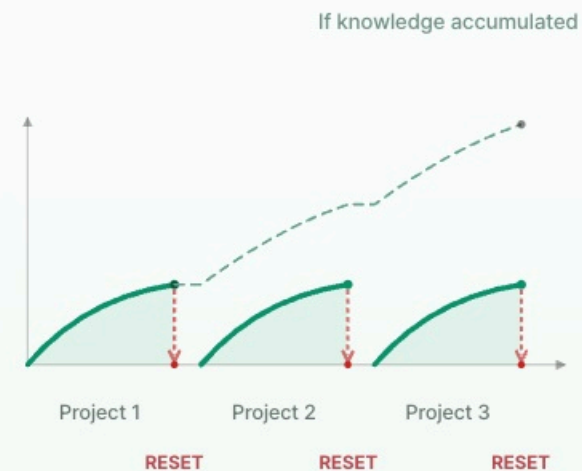
The search space explodes.



Too many choices. Too little time.
Good decisions require search.

03 THE KNOWLEDGE GAP

Know-how resets with every project.

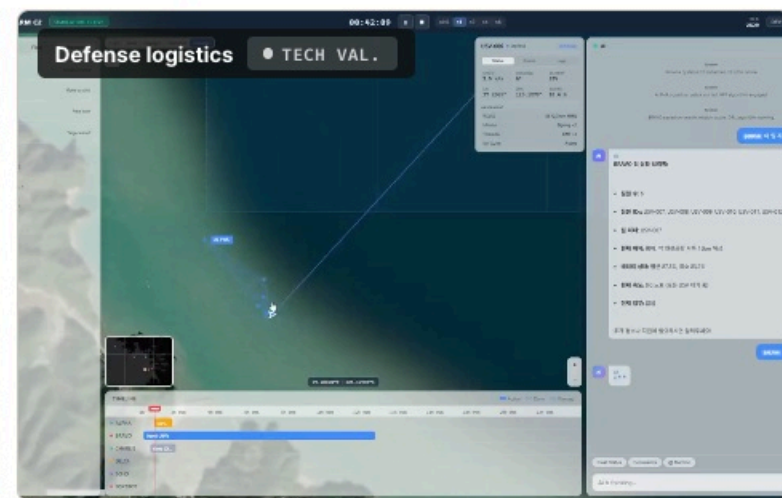
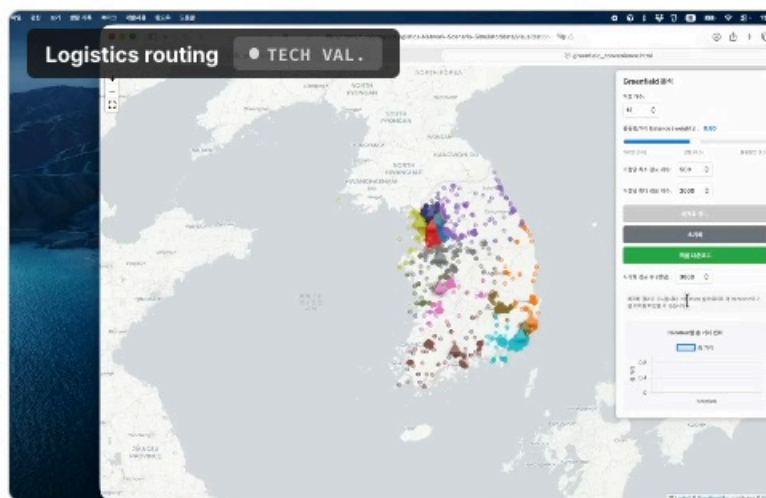
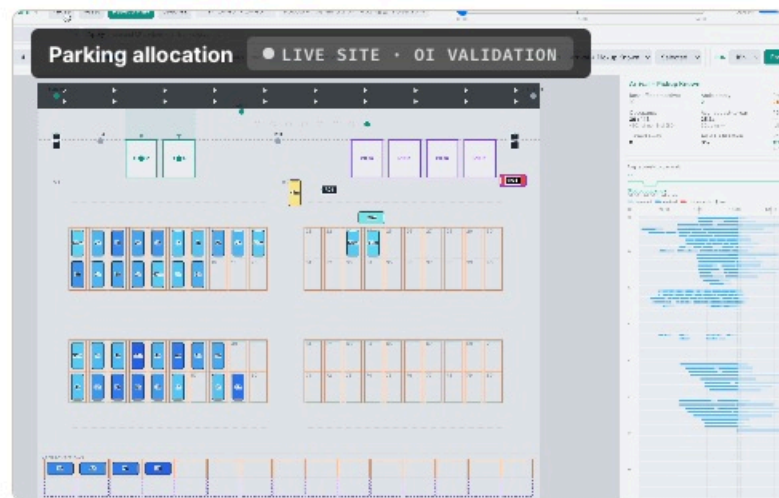
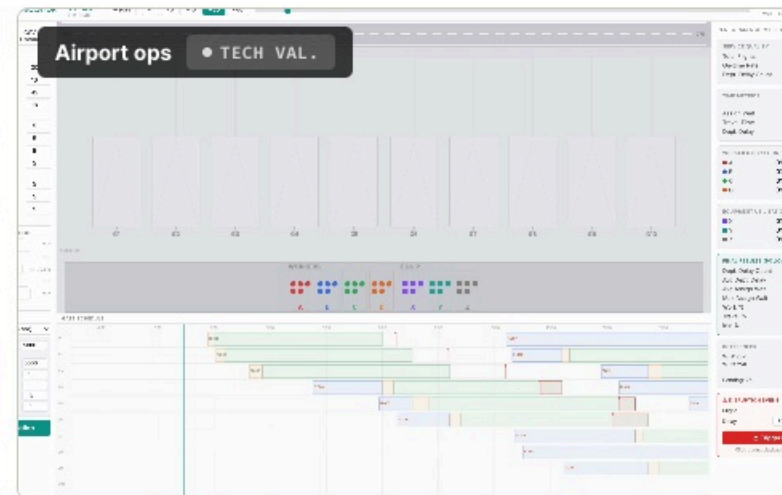
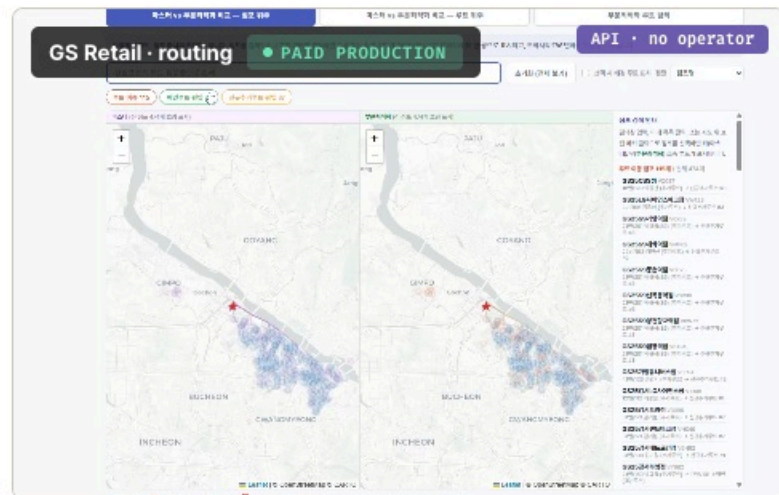


A problem solved. A lesson lost.
The next project starts again.

• PROOF FIRST • ALREADY DEPLOYED

Already Delivering Operational Intelligence Across Industries

Different industries, different OCP capabilities, one production and operating architecture. 5 paid production deployments.



LOGISTICS – 20%+ delivery cost reduction

HEALTHCARE – planning cycle 2 wk → 90 sec

5 paid deployments in total

• Paid production • Technical validation

● GPU-PARALLEL LEARNED OPTIMIZATION • BUILT AND BENCHMARKED BY OMELET RESEARCHERS

We Build the Intelligence Behind the Factory

Generation → Test-Time Search → Fine-Tuning with Human Feedback

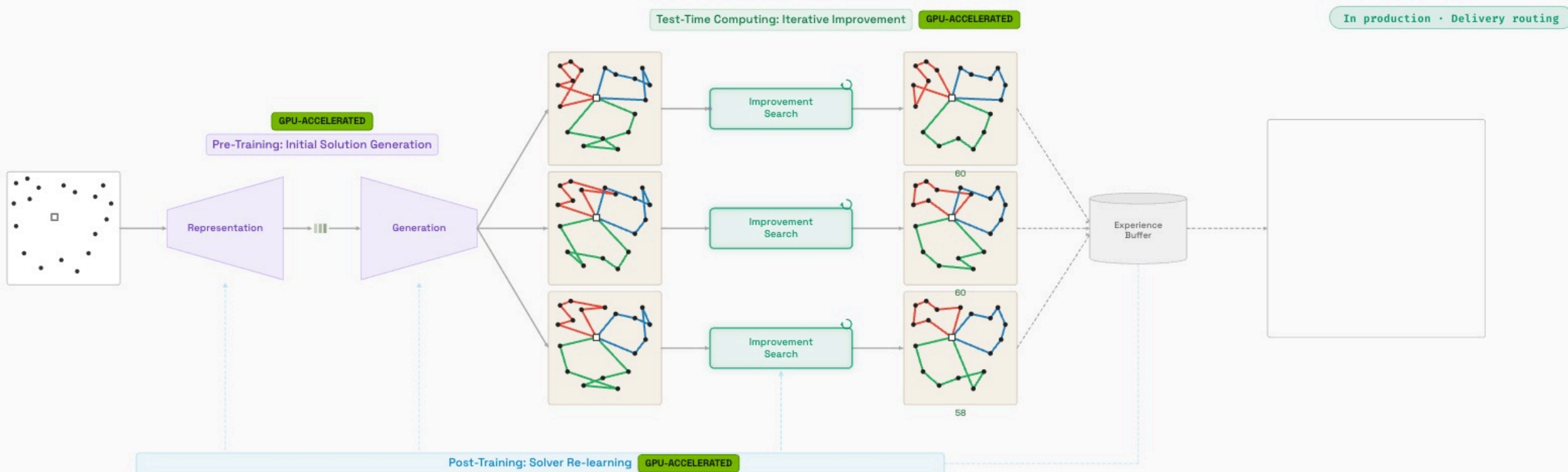
ALIGNED WITH NVIDIA'S THREE SCALING LAWS • GTC 2025 – EVERY STAGE CONSUMES GPU

RL4CO • KDD 2025 ORAL 27 CO environments • 23 state-of-the-art baselines

ROUTING

ROBOT

CHIP



● FROM FOUNDATION MODEL TO DEPLOYABLE APPLICATIONS

One Intelligence Factory, Many Industrial Applications

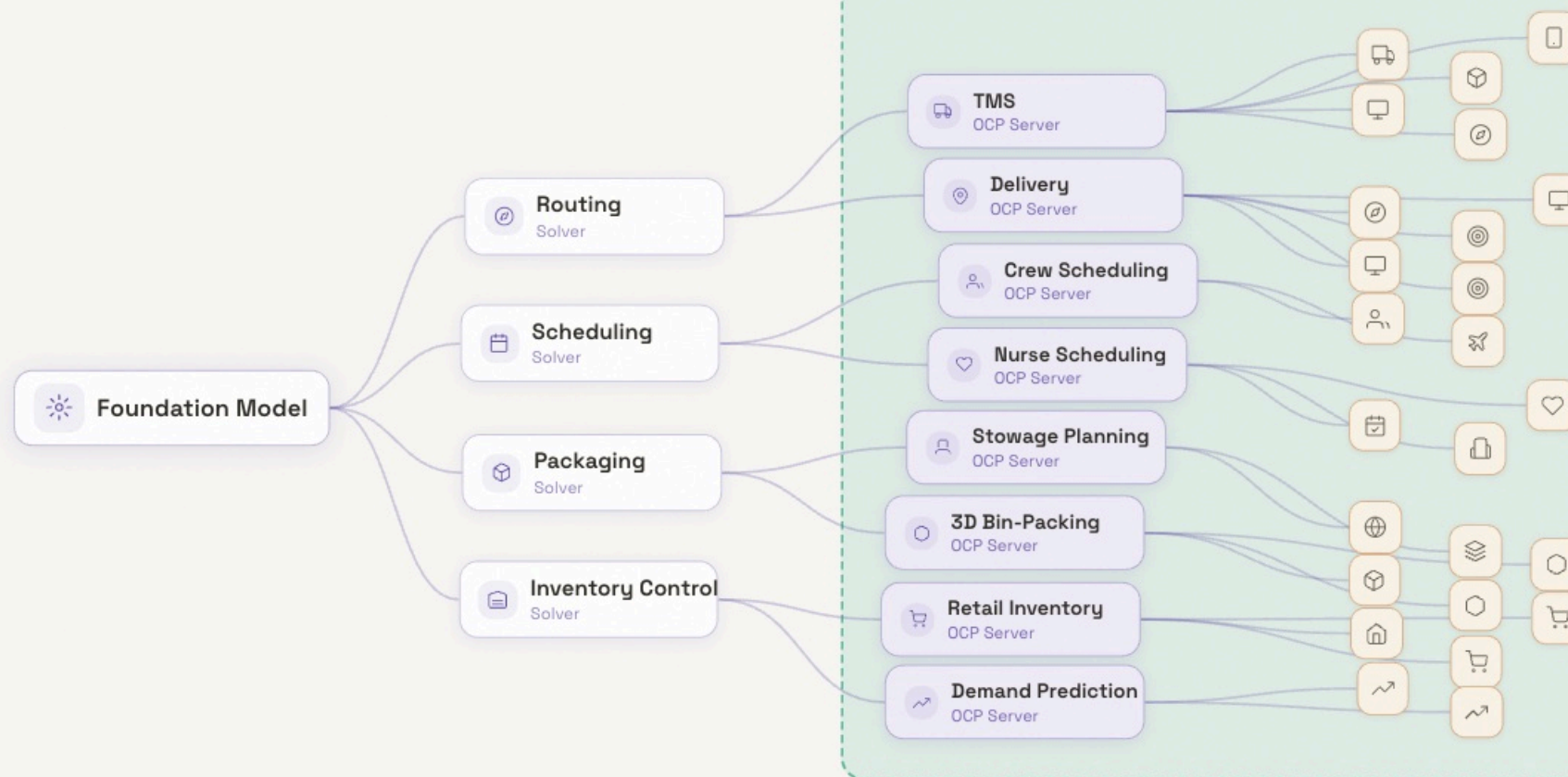
Foundation creates the intelligence • OCP Server makes it reusable • OIP makes it operable.

OPTIMIZATION AI
FOUNDATION MODEL

PROBLEM-CLASS
SOLVER APIS

DOMAIN-SPECIFIC
OCP SERVERS

OI
APPLICATIONS



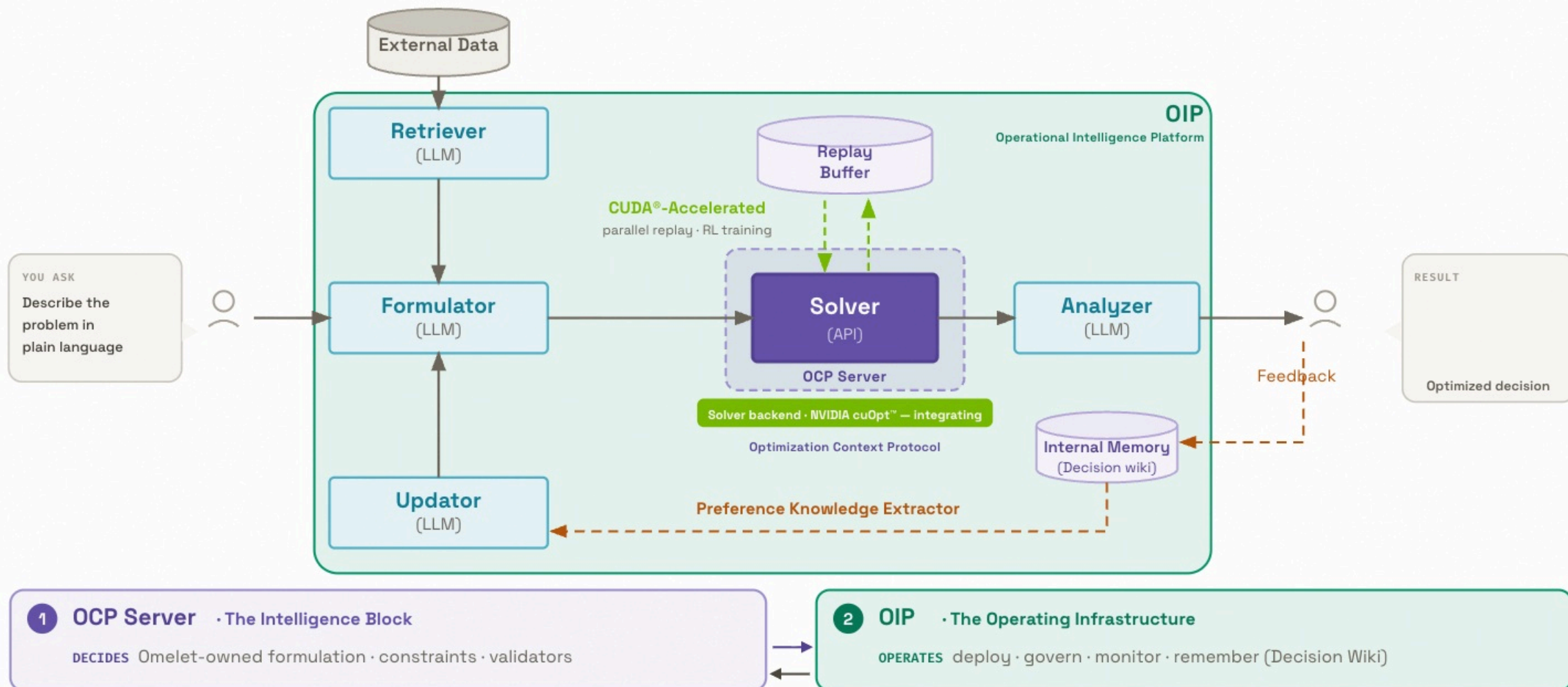
OIP
Operational Intelligence Platform
Deploy • Connect • Monitor • Govern
Secure • Evaluate • Remember • Improve

One platform •
every OCP •
one substrate

● APPLICATION LAYER • INTENT TO CONSTRAINT-VALID DECISION

OCP Provides the Intelligence. OIP Makes It Operable.

OCP is the intelligence block that decides; OIP deploys, governs, monitors and improves every OCP.



● THE PRODUCTION SYSTEM • HOW THE TECHNICAL CORE REACHES EVERY CUSTOMER

One Production System: Solve · Accumulate · Deliver

The Factory solves problems from any industry and layers the capability; Studio and Foundry deliver it on OIP.

PHASE 1 • ACCUMULATE

Solve real problems — layer the capability as OCP Servers & Application Kits

PHASE 2 • DELIVER

Studio explores, Foundry assembles the piles into each customer's own app on OIP

CUSTOMER PROBLEMS • ANY INDUSTRY

DEFENSE

Munitions resupply routing

MANUFACTURING

Line scheduling

HEALTHCARE

Nurse rostering

LOGISTICS

Vessel & vehicle stowage

RETAIL

Store delivery routing

TRANSPORT

Airport gate assignment

FINANCE

Cash-in-transit dispatch

ROBOTICS

Robot parking allocation

PRODUCT 01 • CREATE

OI Factory

Turns one customer's decision problem into a validated capability.

- Interview**
one customer meeting — AI drafts the decision wiki
- Formulate**
objective · constraints · data contract
- Solve**
candidate engines — 0 hard violations · KPI ≥ baseline
- Validate**
rebuilt & verified — by another engineer or an agent

DEPOSITS

DEPOSITS

ASSET 01 • ACCUMULATES OCP Servers

Allocation	OCP 06
Dispatch	OCP 05
Assignment	OCP 04
Stowage	OCP 03
Scheduling	OCP 02
Routing	OCP 01
12 BUILT • 6 OPERATING	

EXPLORES

ASSEMBLES

ASSET 02 • ACCUMULATES Application Kits

Branch Kit	KIT 06
Airport Kit	KIT 05
Fleet Kit	KIT 04
Stowage Kit	KIT 03
Nurse Kit	KIT 02
Line Kit	KIT 01
6 KITS	

ASSEMBLES

PRODUCT 02 • EXPLORE

OI Studio

Explore any operational problem on the accumulated OCP Servers, in natural language.

ASK "Can we cut nurse overtime by 15%?" → SCHEDULING OCP

PRODUCT 03 • DELIVER

OI Foundry

Assembles each customer's own application from an OCP Server + an Application Kit + the customer's needs, and ships it on OIP.

SAME PARTS

Hospital B

=

OCP SERVER

Scheduling

+

APP KIT

Nurse Kit

+

CUSTOMER NEEDS

union rules · night float

12 SHIPPED → DOZENS MORE, SAME FILES

ASSET 03 • OPERATES

OIP

Operational Intelligence Platform

DEPLOY

roll-out

✓

GOVERN

policy · audit

✓

MONITOR

KPIs · drift

✓

REMEMBER

decision memory

✓

→ CUSTOMER
Hospital B

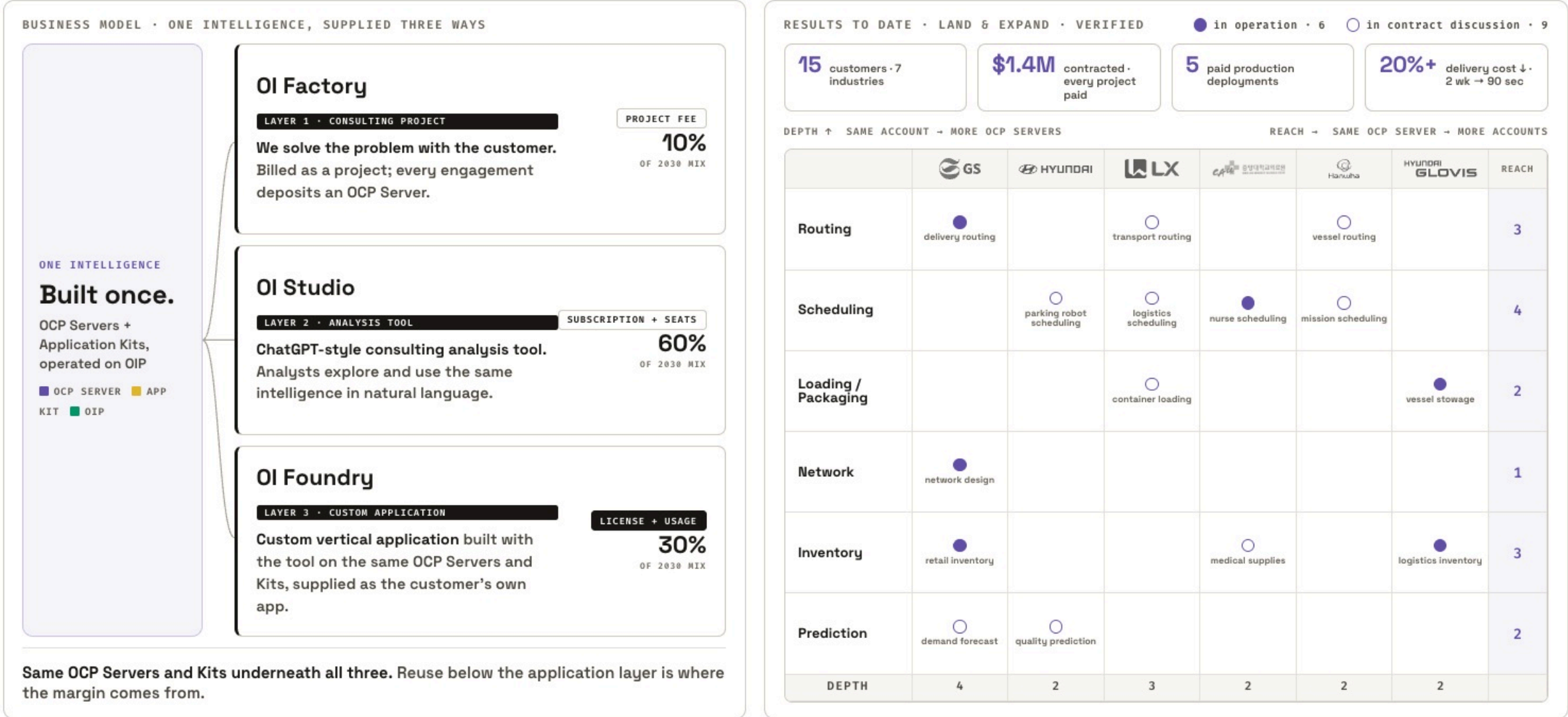
THE LOOP

Every solved problem adds a layer. The same piles assemble dozens of applications.

● BUSINESS • HOW THE PRODUCTION SYSTEM EARNS, AND WHAT IT HAS PRODUCED SO FAR

Projects Deposit OCPs. OCPs Sell Again.

One intelligence, supplied three ways: as a consulting project, as an analysis tool, and as a custom application.



● HOW OMELET WORKS WITH NVIDIA • THE CUOPT™ INTEGRATION PATH

NVIDIA cuOpt™ as a GPU Backend Behind the OCP Contract

VRP first, then LP/MIP. Same problems, backend swapped, measured end to end — a complementary solver stack, not a replacement.

● STATUS **CUDA® INTEGRATING** **CUOPT™ INTEGRATING • VRP FIRST** **LP / MIP NEXT** **ISAAC SIM™ FUTURE**

Not in production yet: the integration structure and benchmark scenarios are being defined.

END-TO-END, NOT STANDALONE • cuOpt™ IS ONE GPU BACKEND BEHIND THE SAME OCP CONTRACT



NOW

VRP first

Existing OCP routing scenarios, **current solver vs cuOpt™**, nothing else changed. Focus on the large, parallel problems where GPUs should win.

NEXT

LP / MIP, then selection

Extend the cuOpt™ backend to LP/MIP and let the OCP choose per problem. **Omelet engines + cuOpt™ = a complementary stack**, not a replacement.

WITH NVIDIA

Co-designed evaluation

Select benchmarks and problem scales **with the cuOpt team**; run large-scale evaluation and performance tuning on NVIDIA GPUs; ship the end-to-end demo.

THE ASK

A GPU environment to evaluate cuOpt™ at scale • Benchmark & scale co-selection with the cuOpt™ team • Reference integration & customer-scale demo



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Operational Intelligence_

OMELET × NVIDIA

INCEPTION GRAND CHALLENGE · FINAL

Omelet industrializes
Operational Intelligence.
NVIDIA accelerates it.



ONE NVIDIA INTEGRATION → MANY ENTERPRISE DECISION WORKLOADS

solve → accumulate → deliver → learn → every project makes the next one faster

info@omelet.ai

OMELET.AI · SEOUL

APPENDIX

Business & Team

Market · Competition · Moat · Revenue Model · Founder & Team

● MEASURED ENGINE BENCHMARKS • SAME INSTANCES • SAME HARDWARE • NAMED SOLVERS

The Right Engine for Every Decision — Measured

No single engine wins every problem class — we benchmark Omelet engines, classical solvers and NVIDIA cuOpt™ head-to-head on our workloads.

WHERE NVIDIA CUOPT™ WINS • ON OUR WORKLOADS

cuOpt™ 26.08 • CUDA® 13 • single GPU

WAREHOUSE PLANNING • GPU MILP • 317K VARIABLES

TIME TO OPTIMUM ↓



ONLY SUBLINEAR SCALING — 5.1× VARS → 1.7× TIME (others 6.3–9.4×) • same optimum, all solvers

CUSTOMER ROUTING • CVRPTW • GS RETAIL • 537–586 ORDERS

GAP @ 60 s ↓



cuOpt™ best distance @60 s: 2,610 km • 2/3 wins at both budgets • RTX 4090 / A6000 • repeated runs, re-validated

OMELET ENGINE AT THE FRONTIER • CVRPTW

Solomon • 56 instances • 10 s limit



CUSTOMER SCALE • GS RETAIL Omelet & cuOpt™ statistically tied at the frontier ($\Delta < \text{run-to-run } \sigma$) • both far ahead — OR-Tools +9.5%

*LKH-3 feasible on 52/56 • stochastic engines vary ± 0.5 –1.3%p between runs • gap ↓ lower is better
Benchmark infrastructure: RL4CO (KDD 2025 Oral) • learned-hybrid up to ~250× vs LKH-3 at 7K nodes (Table 5.4)

● FULL ENGINE-BENCHMARK RESULTS • ENVIRONMENT & VERIFICATION

Engine Benchmarks — Full Results

WAREHOUSE PLANNING • GPU MILP • 316,814 VARS / 240,120 CONSTRAINTS

SOLVER	HW	SOLVE	TOTAL	VS cuOpt™
NVIDIA cuOpt™ 26.08	GPU×1	13.6 s	15.4 s	baseline
SCIP 9.0	CPU 64T	18.1 s	21.4 s	+1.39×
CP-SAT 9.11	CPU 64T	15.0 s	22.6 s	+1.47×
CBC 2.10	CPU 64T	40.8 s	44.1 s	+2.86×

Scaling 62K → 317K vars (5.1×): cuOpt™ ×1.7 • CP-SAT ×6.3 • SCIP ×7.9 • CBC ×9.4 — at single-warehouse scale CPU solvers lead (SCIP 0.2–2.7 s); the crossover is scale.

All four solvers reach the same optimum (1,139 PLT • –24.3% peak space). No mid-run incumbents — time-to-optimal is the operative metric; cuOpt™ earliest at every budget that yields a solution.

Xeon Gold 6326 ×2 (64T) • RTX 4090 (2.0 / 24 GiB used • FP64-limited consumer GPU) • CUDA 13.0 • Ubuntu 24.04 • wall-clock, 3-run median • solutions re-simulated independently: 0 violations

CVRPTW • SOLOMON (56 × 100-CUSTOMER • 10 S) & GS RETAIL (537–586 ORDERS)

METHOD	HW	GAP ↓	WINS	TIME
OMELET RouteOpt	CPU	+0.08%	38/56	10.0 s
NVIDIA cuOpt™	CPU+GPU	+1.76%	13/56	9.7 s
OMELET RouteOpt • ALNS config	CPU	+2.53%	3/56	8.1 s
LKH-3*	CPU	+3.41%	2/56	47.6 s
OR-Tools	CPU	+4.56%	0/56	10.0 s

GS RETAIL • AVG KM	10 s KM	10 s GAP	60 s KM	60 s GAP
NVIDIA cuOpt™	2,684	+1.09%	2,610	+0.83%
OMELET RouteOpt	2,676	+1.37%	2,622	+1.70%
OMELET RouteOpt • ALNS config	2,984	+12.22%	2,766	+7.16%
OR-Tools	3,000	+12.79%	2,815	+9.50%

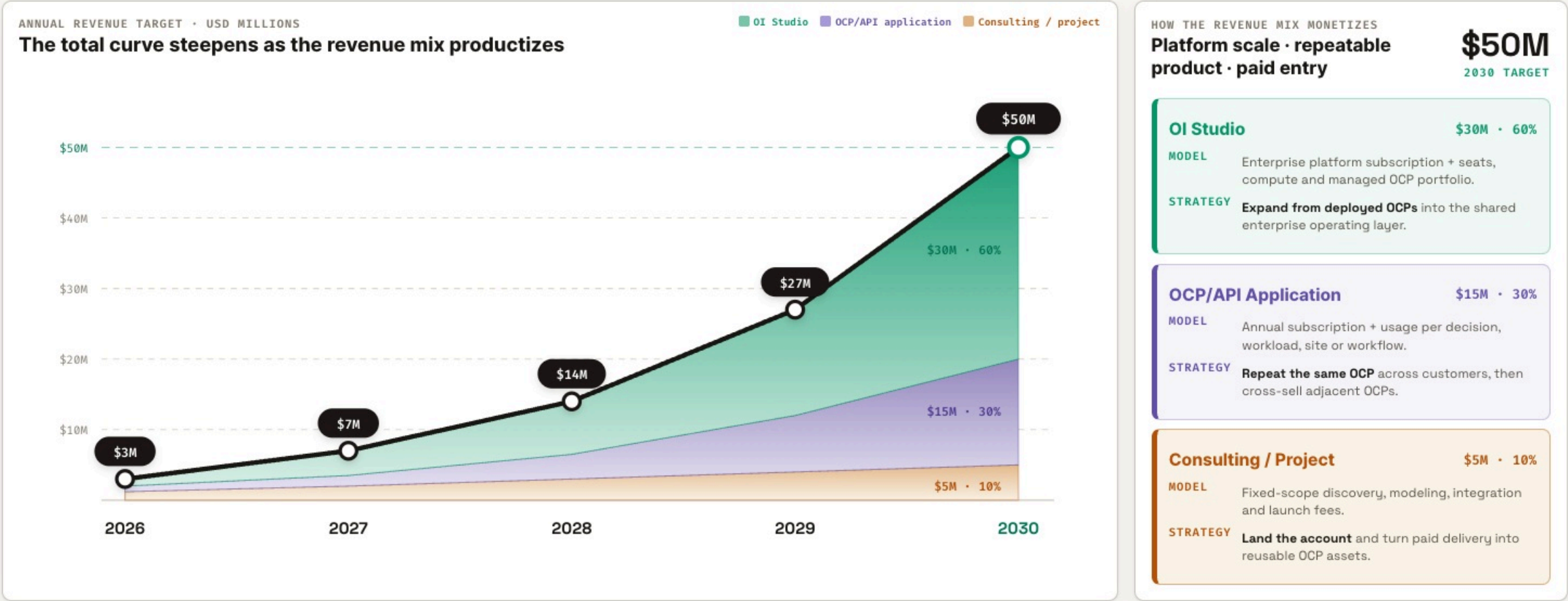
*LKH-3: feasible on 52/56 Solomon; excluded at GS Retail scale — no feasible solution at 1,830 s (30× budget). Gap = vs best across solvers per instance (moving reference — compare km across budgets, not gaps). Stochastic σ between runs: cuOpt™ ±1.24 / ±0.50%p • RouteOpt-ALNS ±0.87 / ±1.27%p — GS Retail Δ is within noise.

3 instances • A3 replenishment scenario • objective: drive distance (55 km/h) • RTX A6000 ×1 • Aug 2026

● WHAT HAPPENS IF THE REPEATABILITY TEST WORKS?

Repeatability Converts Services Into Productized OI

Management case: \$3M in 2026 → \$50M in 2030, with productized OI rising to 90% of revenue. TARGET SCENARIO • NOT AUDITED ACTUALS



Source: management-case revenue model in Omelet_IR.html. 2026–2030 targets, not audited actuals. Original totals are preserved; product categories are aligned to the current OI Factory commercial model.

MANAGEMENT CASE

The \$50M curve is the financial output of repeatability, expansion and product mix — Series A must prove the assumptions underneath it.

● FOUNDER & TEAM · LAB RESEARCH BECOMING COMMERCIAL SOFTWARE

Ten Years of Lab Research, Industrialized by One Execution Team

28 people · KAIST-anchored optimization talent + industry builders



Jinkyoo Park FOUNDER & CEO

Stanford PhD · KAIST IE / AI Professor · 10 years

RESEARCHER 61 ML papers · 36 at NeurIPS / ICML / ICLR · RL4CO 897 stars — extended by MILA · Stanford · MIT CSAIL

ENTREPRENEUR Founded Omelet (2023) — 12 OCP Servers built · 6 in operation · \$1.4M contracted

OPERATION TEAM

Jongbum HEAD

Head of Management · Ph.D. KAIST

Eunji

Management Manager

Jina

Management Manager

BUSINESS DEVELOPER

Youngseok VP

Head of Biz Dev.
CJ Logistics · LogisALL

Suhyun

Biz Development
B.A., HUPS

First US GTM hire (PLANNED)
funded by this round · US commercial lead

PROBLEM SOLVER

Changhyun Kwon CO-FOUNDER & CTO

KAIST IE Prof. · Ph.D. Penn State

Kyuree

Combinatorial Opt.
Ph.D. IE, KAIST

Jiwoo

Combinatorial Opt.
M.S. IE, KAIST

Huijie

Multi-Agent Pathfinding
M.S. IE, KAIST

Jihwan

Search-Based Opt. · MLOps
Ph.D. IE, KAIST

Seayoung

Optimization Eng.
B.S. IE, SNU

Hyoyun

Optimization Eng.
B.S. IE, SNU

Jaebeom

Optimization Eng.
B.S. IE, SNU

Geonhyoung

Optimization Eng.
B.S. IE, SNU

Haewon

Optimization Eng.
B.S. IE, KAIST

PRODUCT BUILDER

Seounghoon CPO

AI Strategy · Ph.D. CS, KAIST

Seunghyun

OR Eng.
Ph.D. Math, SNU

Jongsun

Platform Dev.
10yr · VROONG · LG

Wonjun

Backend · Java
ABLY · VROONG

Meeso

Frontend Systems
Wesang (Yogiyo)

Youngjin

Cloud · AI Backend
UC Berkeley · KT Cloud

Seayoung

Backend · DevOps
SNU · MEGAZONE

Jieun

Product Design
M.S. SCAD · Lenme (SF)

Juni

Optimization AI Agent
B.S. EE, Korea Univ.

Jiyeon

Product Dev.
B.S. IE, SNU

Jungho

QA Engineer

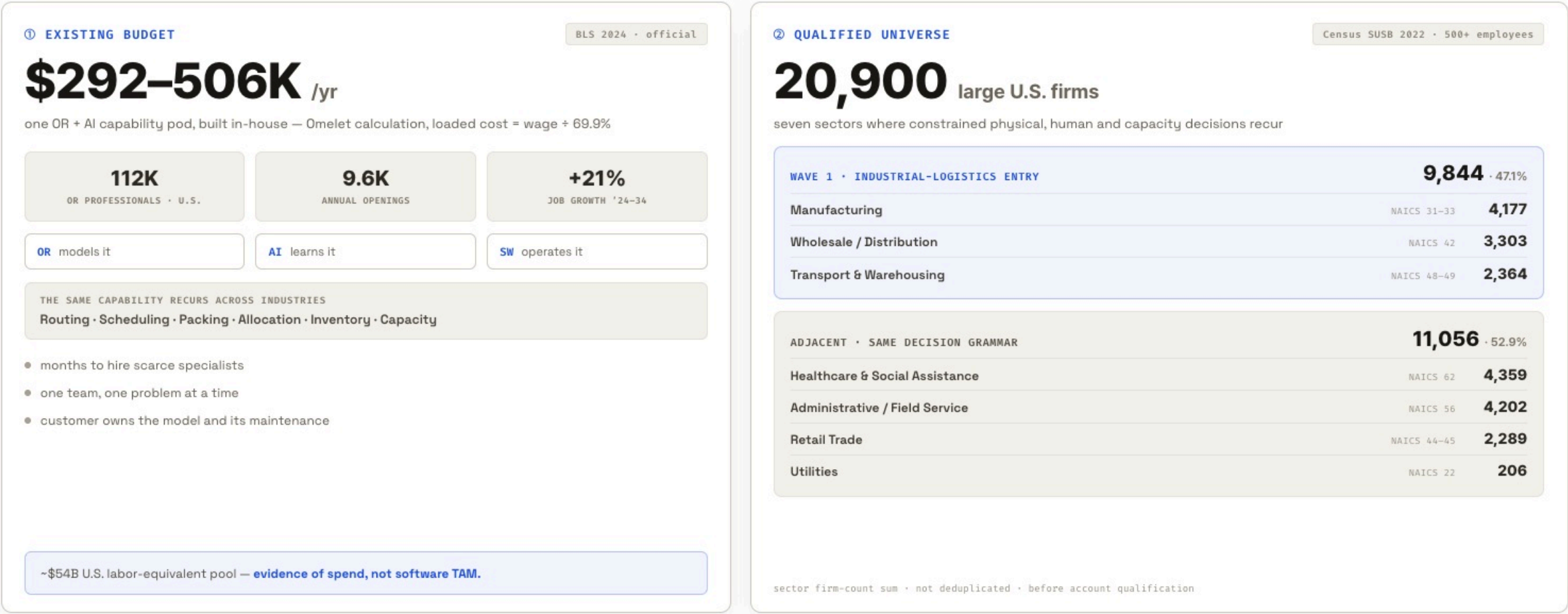
THREE ROLES

Problem Solvers formulate. Product Builders industrialize. Business Developers scale distribution.

● IS THERE ALREADY A BUDGET — AND HOW MANY BUYERS?

The Budget Already Exists — Across an Enumerable Universe

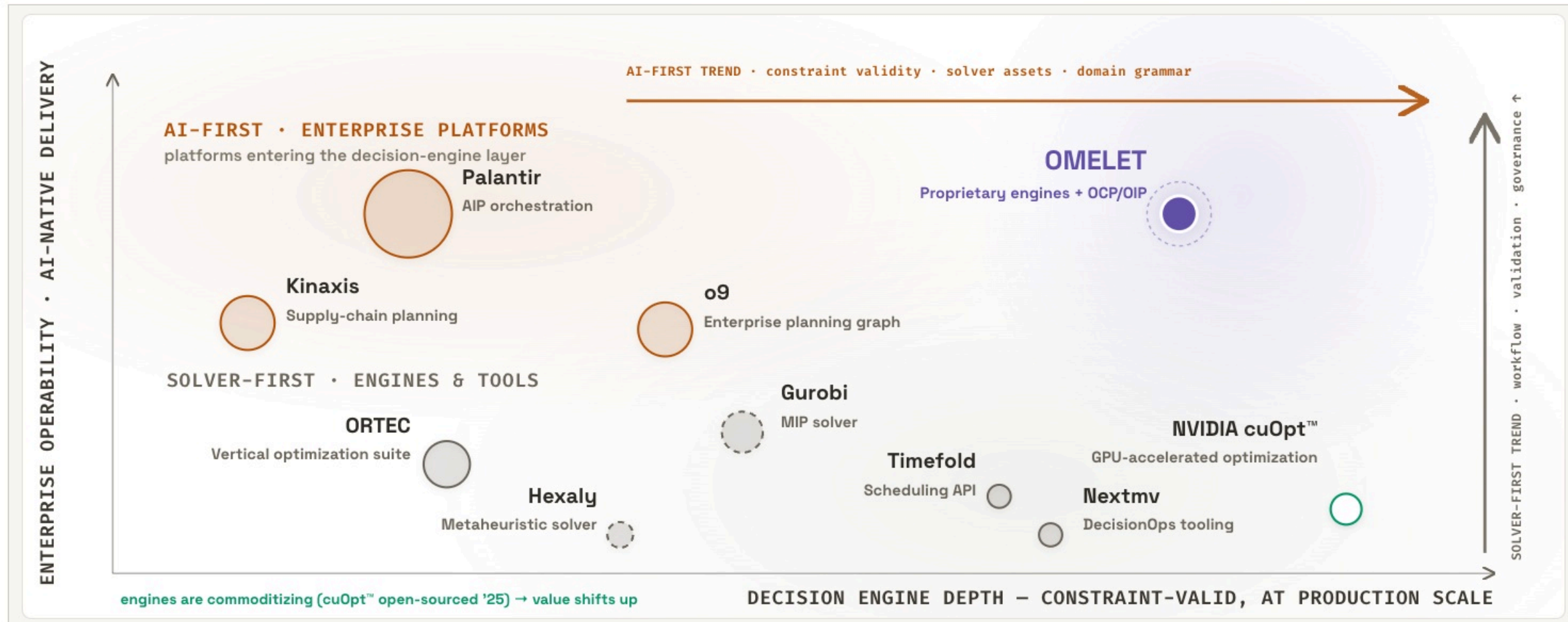
Enterprises already pay for scarce optimization capability as labor; 20,900 large U.S. firms across seven sectors form the screening base. Bottom-up, official data only (BLS • Census).



● WHO COMPETES — AND FROM WHICH DIRECTION?

Omelet Unifies Solver Depth With Enterprise Operability

We do not replace systems of record like the TMS. We replace the bespoke optimization build — the capability layer both camps are converging toward.



POSITION

Engines are moving up. Platforms are moving down. Omelet already operates at the convergence — in paid production.

● COMPOUNDING ADVANTAGE • EVERY DEPLOYMENT ADDS ASSETS

Every Deployment Deepens the Moat

Configured, not rebuilt • customer-controlled, not pooled — the delivery loop's output is the moat: each cycle deposits assets a competitor cannot shortcut.

